

AI-Powered Crime Scene Camera Detector

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Introduction & Operational Context

When law enforcement officers arrive at a crime scene, one of their most critical immediate tasks is identifying civilian surveillance footage that may have captured the incident. Historically, this operation has been handled completely manually, forcing investigators to walk perimeters, scan structural facades, and knock on doors. This exhaustive process frequently takes hours, directly draining the crucial "golden hours" of an investigation where evidence preservation is most vital and when the risk of automatic video overwriting is highest. To eliminate this operational bottleneck, this project introduces an automated computer vision pipeline capable of detecting civilian cameras in imagery. By immediately highlighting camera locations, the system compresses hours of manual surveying into mere minutes, allowing field units to secure decisive evidence instantly and redirect their focus toward the core investigation.

Data Construction & Augmentation

To ensure the foundational reliability of this detection architecture, I developed a diverse, custom-built dataset designed to reflect complex real-world conditions.

- Initial Curation: Gathered 5,074 images (positive camera and negative non-camera) featuring urban backgrounds, people, food, and complex environments.
- Augmentation Strategy: Applied perspective transformations, digital noise injection, rotation, horizontal flipping, and grayscale conversions to maximize generalization and mitigate overfitting.
- Final Volume & Split: Expanded to a balanced 27,000 images, divided into 70% training, 20% validation, and 10% testing.



Examples of Pictures in the dataset

Methodology & Pipeline Design

- Parallel Training: Trained two independent deep learning pipelines simultaneously to evaluate their real-world suitability.
- Pipeline 1 (Local): Fine-tuned a locally configured YOLOv8 Medium architecture, optimizing its feature extraction backbone and decoupled classification head.
- Pipeline 2 (Cloud): Applied transfer learning using a ResNet50 classification model hosted on the cloud-managed Roboflow platform.
- Comparative Baseline: Both networks were optimized to strictly compare a custom standalone framework against a platform-managed training environment.

Results & Evaluation

- Evaluation divergence: Evaluating both systems revealed a fascinating divergence between controlled test datasets and actual field conditions.
- ResNet50 baseline: In the automated dataset evaluation across 2,700 benchmark images, the Roboflow ResNet50 model initially emerged as the clear leader on paper.
- ResNet50 metrics: The model achieved an outstanding 98.11% accuracy, 97.51% precision, and 98.74% recall.
- YOLOv8 Medium baseline: In comparison, the YOLOv8 Medium configuration demonstrated a more moderate dataset performance.
- YOLOv8 Medium metrics: The model recorded a balanced 82.0% accuracy, 84.9% precision, and 82.0% recall.

Model Pipeline	Accuracy	Precision	Recall	F1 Score
Yolov8 Medium	82.0%	84.9%	82.0%	81.6%
ResNet50	98.1%	97.5%	98.7%	98.1%

Compared Evaluation of the two models

Empirical Performance Divergence

Evaluating both systems revealed an important contradiction between controlled test datasets and actual field conditions. In the automated validation across 2,700 benchmark images, the cloud-managed ResNet50 model initially appeared superior, achieving an outstanding 98.11% accuracy, 97.51% precision, and 98.74% recall. In comparison, the custom YOLOv8 Medium configuration recorded a more moderate baseline performance, demonstrating a balanced 82.0% accuracy, 84.9% precision, and 82.0% recall. Relying strictly on these automated metrics would suggest that the platform-managed solution is the definitive choice for deployment.

Real-World Manual Testing

However, the performance dynamics reversed completely when both models were subjected to out-of-distribution manual testing using diverse, ambiguous real-world inputs. The YOLOv8 Medium model achieved a perfect 12 out of 12 correct classifications, demonstrating robust internal feature representation and making high-confidence negative predictions on non-camera objects. Conversely, the ResNet50 model dropped to 10 out of 12 due to critical false positives. It incorrectly flagged atypical test subjects, such as a vehicle and a food bucket, as surveillance cameras due to spurious correlation of visual lines, proving highly sensitive to environmental noise.

Test	Subject	Roboflow Confidence	YOLOv8 Confidence
Test 1	Ronaldo	77.7%	100.0%
Test 2	Camera	99.7%	100.0%
Test 3	Cat	69.5%	100.0%
Test 4	Dog	53.5%	100.0%
Test 5	Camera	99.6%	100.0%
Test 6	Messi	99.4%	100.0%
Test 7	Elon Musk	98.8%	100.0%
Test 8	Car	59.7%	100.0%
Test 9	KFC Bucket	51.5%	89.0%
Test 10	Hallway	98.4%	99.5%
Test 11	Camera 2	99.6%	99.8%
Test 12	Camera 3	99.6%	100.0%

Comparison of the Manual Testing of the models

Operational Suitability & Conclusion

For tactical law enforcement deployment, structural resilience against false alarms is paramount to maintain officer trust and prevent wasted operational hours. While the platform-managed architecture holds a statistical advantage within its precise dataset distribution, YOLOv8 Medium stands out as the operationally superior model for field enforcement. The definitive edge for YOLOv8 Medium lies in its deployment flexibility; it completely bypasses vendor platform restrictions and easily exports via ONNX frameworks. This architecture provides a highly reliable, independent edge-computing solution capable of processing field data locally without external network dependencies.

Technical Roadmap & Future Work

The engineering development roadmap focuses on transitioning the pipeline from basic image classification to full-scale object detection. Moving YOLOv8 into its native detection mode will allow the model to draw precise bounding boxes around every identified camera, providing field officers with exact coordinates and enabling the system to count multiple obscured devices simultaneously. Furthermore, upcoming iterations will integrate live Geographic Information System (GIS) mapping to plot detected cameras automatically onto a spatial overview of the crime scene. Advanced optimization will also focus on processing live video streams from body cams or drones and reducing model weight to run the network locally on rugged field tablets without an active internet connection.